

SLIDE 1: Title Hook: MPLADS-SATARK ()

Tagline: Automated AI Risk Scoring Multi-Layer Fraud Detection Explainable Auditing

KEY BULLET POINTS FOR SLIDE:

- * Problem Statement 26102 Ministry of Statistics Programme Implementation (MoSPI DIID)
- * What is MPLADS Over Rs 4,000 Crore is given annually to 779 MPs across 788 districts for local public infrastructure.
- * The Core Hook: Official portals only show WHAT happened in the past. They do not warn officers BEFORE public money is wasted on duplicate works, overpriced tenders, or stalled projects.
- * Our Solution: SATARK () an AI-powered proactive vigilance system that scans government works in real-time, catches corruption patterns, and explains every red flag in plain language.

WHAT TO SAY TO JUDGES:

"Start with high energy: Good morning respected jury members Every year, 4,000+ Crore of taxpayers money is spent on MPLADS for roads, schools, and drinking water. But today, audits happen 2 years too late. We present MPLADS-SATARK an AI system that catches fraud, overpricing, and ghost projects before public funds are wasted "

SLIDE 2: Ground Realities 4 Massive Problems in MPLADS

Tagline: Why Government Audits Fail Today and Where Public Money is Leaking

KEY BULLET POINTS FOR SLIDE:

- * 1. Double Invoicing Ghost Works: The same rural road or hall is funded under MPLADS AND also billed under MGNREGA or PMGSY.
- * 2. Inflated Estimates (Overpricing): Contractors charge Rs 50 Lakhs for a basic road that CPWD standard rates benchmark at Rs 20 Lakhs.
- * 3. Contractor-Agency Cartels (Favoritism): 70 to 80 of an MPs funds repeatedly awarded to a single favorite agency or contractor.
- * 4. Statutory Violations March Rush: Sanctioning ineligible works (places of) ' (worship, private clubs) and dumping 30 of
- * 5. Zombie Projects: Works that are 10 complete on paper, stalled for 3 years, with zero progress.

WHAT TO SAY TO JUDGES:

"Highlight ground reality: Judges, fraud in public works doesnt happen with complicated equations. It happens when someone claims money for the same road twice, charges double the CPWD rate book, or creates a fake bill of 4.98 Lakhs to avoid mandatory online tenders "

SLIDE 3: Our Solution: MPLADS-SATARK Architecture in Simple Words

Tagline: How We Catch Irregularities Without Black-Box Confusion

KEY BULLET POINTS FOR SLIDE:

- * 100 Real Government Data: Sourced directly from MoSPI eSAKSHI live endpoints (176,831 real works, 109,475 payment vouchers, 3636 States UTs).
- * Multi-Layer Defense (6 Independent Signals): A project is never flagged based on one single model. It requires multiple red flags to become high priority.
- * Honest Data Provenance Badges: Every number carries a clear badge REAL, DERIVED, ESTIMATED, or STATUTORY. Zero fake numbers, zero black boxes.
- * Explain-My-Score Waterfall: Instead of an obscure decimal, officers see plain English points: e.g. +30 CPWD Overpricing, +25 Contractor Favoritism, +20 March Rush.

WHAT TO SAY TO JUDGES:

"Explain clearly: We don't use black-box AI. Government officers cannot take disciplinary action on a black-box percentage. SATARK gives an itemized additive waterfall: why was this project flagged, which rule was broken, and how much is the cost variance "

SLIDE 4: The 6 Independent AI Detection Signals (Explained Simply)

Tagline: Each Signal Catches a Different Type of Financial or Operational Fraud

KEY BULLET POINTS FOR SLIDE:

- * Signal 1 (Cost Delay Outliers): Uses Isolation Forest ML calibrated against official CPWD Delhi Schedule of Rates (DSR) to spot inflated estimates.
- * Signal 2 (Contractor Cartels Network Graph): Measures fund concentration (HHI) and spots syndicate bidding where 1 contractor monopolizes funds.
- * Signal 3 (Statutory Compliance Engine): Policy-as-Data checking mandatory 15 SC 7.5 ST quotas, and 7 Negative List clauses (e.g. prohibited religious) ' (structures).
- * Signal 4 (Cross-Scheme Deduplication): Probabilistic matching comparing MPLADS works against ISRO Bhuvan MGNREGA registries in the same Gram Panchayat.
- * Signal 5 (Benfords Law Forensic Accounting): Analyzes leading digits (1-9) in bills to detect human-padded figures and 4.98L split tenders.
- * Signal 6 (Cox Survival Hazard for Zombie Projects): Statistical time-to-event model forecasting completion curves and warning of stalls 6 months early.

WHAT TO SAY TO JUDGES:

"Walk through the 6 signals: Signal 1 checks pricing. Signal 2 checks who got the contract. Signal 3 checks government guidelines. Signal 4 checks if another scheme already built it. Signal 5 checks the mathematics of the invoice. Signal 6 predicts if the project will die before finish "

SLIDE 5: Bifurcated Architecture: Cold Path vs Hot Path Live Demo

Tagline: Engineered for Government Scale and Flawless Live Demonstration

KEY BULLET POINTS FOR SLIDE:

- * Cold Path (Offline Analytics): Indexes 176,831 projects in under 2 seconds for instant nationalstate aggregations and filtering.
- * Hot Path (Live Stage Injection): Live scoring of any newly proposed work in UNDER 15 MILLISECONDS (against 50ms SLA).
- * Focus Context Ego-Graph: Rather than crashing browser with 27,000 nodes, renders 1-2 hop neighborhood (10-15 related nodes) on click.
- * Bitemporal Policy Engine: Versioned rules ensure projects sanctioned under older guidelines are evaluated against the rules active on that date.

WHAT TO SAY TO JUDGES:

"Emphasize live demo speed: Most teams struggle with live demo lag. We built a bifurcated architecture where live injection scores any new project in 1 millisecond on stage "

SLIDE 6: 4 Role-Based Dashboards for Every Level of Governance

Tagline: One Unified Source of Truth Scoped to 4 Key Administrative Roles

KEY BULLET POINTS FOR SLIDE:

- * 1. Union Ministry DIID: National macro overview, inter-state benchmarking, fund utilization rates, and nationwide GIS risk density map.
- * 2. State Nodal Authority: Inter-district performance roll-up, agency concentration heatmaps, and unspent balances.
- * 3. District Magistrate Collector: Prioritized inspection queue, one-click Explain-My-Score modal, and print-ready field audit dossiers.
- * 4. Honble Member of Parliament: Self-check scorecard, live SC (15) and ST (7.5) statutory quota meters, and pending approvals.

WHAT TO SAY TO JUDGES:

"Demonstrate governance value: Different users need different views. The MP wants a self-check tool to stay compliant. The District Collector wants an inspection list. The Ministry wants national macro trends. SATARK serves all 4 from one single database "

SLIDE 7: Closing the Loop: One-Click Printable Audit Dossier

Tagline: Moving From Digital Flags to Real Administrative Ground Action

KEY BULLET POINTS FOR SLIDE:

- * Official Inspection Memorandum: 1-click printable CAGDistrict Engineer notice with pre-filled citations and evidence.
- * On-Ground Verification Checklist: Standardized checks for physical asset existence, signboard display, and Measurement Book (MB) reconciliation.
- * Human-in-the-Loop Governance: AI never freezes funds or replaces officers it acts as an intelligent assistant providing audit-defensible proof.
- * Digital Sign-Off Audit Trail: District engineers record photos, GPS coordinates, and inspection findings directly onto the system.

WHAT TO SAY TO JUDGES:

"Show practical impact: AI that just shows red badges on a screen is useless in government. Our system outputs an official inspection memorandum that a junior engineer can take directly to the construction site to verify the work "

SLIDE 8: Why MPLADS-SATARK Wins (The Scorecard)

Tagline: Comparing Our Solution Against Typical Hackathon Submissions

KEY BULLET POINTS FOR SLIDE:

- * 1. Data Integrity: Typical teams use fakesynthetic data. We use 100 real government MoSPI, CPWD DSR, and Bhuvan data.
- * 2. AI Breadth: Typical teams use 1 basic ML model. We use 6 independent mathematical and ML layers.
- * 3. Transparency: Typical teams have black-box scores. We have an itemized additive waterfall with legal rule citations.
- * 4. Legal Grounding: Grounded directly in GFR 2017 MPLADS 2023 Guidelines.
- * 5. Ready Working Prototype: Full REST API, Hot Path live sandbox, and interactive UI ready today.

WHAT TO SAY TO JUDGES:

"Conclude with confidence: In conclusion, MPLADS-SATARK is not a toy project with dummy numbers. It is an audit-defensible, legally grounded, and ultra-fast vigilance platform ready for national rollout across India. Thank you "